ANUSHKA BATTE

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SUMMARY

Final-year CS undergrad (CGPA 3.93/4.00) with 2 peer-reviewed ML publications and applied research projects spanning agentic AI, privacy-preserving ML on homomorphic encryption, causal time-series forecasting, and healthcare AI. Targeting ML/AI and Data Science internships, with focus on applied research and healthcare ML.

EDUCATION

Mukesh Patel School of Technology Management and Engineering (NMIMS) Jul 2021 – Present

B.Tech. Integrated in Computer Engineering | Mumbai, India | **CGPA: 3.93/4.00**

Relevant Coursework: Machine Learning, Artificial Intelligence, Image and Video Processing, Data Structures, Database Management Systems, Object Oriented Programming, Probability and Statistics, Human Computer Interaction

PUBLICATIONS

1. "Exploring Machine Learning Models for Schizophrenia Detection: A Systematic Review." *WIREs Data Mining and Knowledge Discovery* (Wiley), 2025. DOI: 10.1002/widm.70048

2. "Impact of Automation on Neuroplasticity: A Systematic Literature Review and Conceptual Framework of Cognitive Adaptation in Human–AI Interaction." *Journal of Cortexplore*, Vol. 1 No. 2, 2026. https://cortexplore.org/index.php/jce/article/view/11

EXPERIENCE

Acuradyne (incubated at SINE, IIT Bombay) Jun 2026 – Present

Data Science Intern | Mumbai, India

- Performing exploratory data analysis (EDA) on ECG signal data using Python and Pandas to support downstream ML model development for cardiac health applications.

PROJECTS

Adversarial ML Against Homomorphic Encryption (CKKS) — Side-Channel Leakage Jan 2026 – Apr 2026

Python, TenSEAL, PyTorch, scikit-learn, Flask, NumPy/Pandas, Matplotlib

- Constructed privacy-preserving ML pipeline benchmarking 4 classifiers (Random Forest, SVM, Torch MLP, baseline) on a 1,000-sample dataset across 5 CKKS encrypted operations, achieving **81% attack accuracy** (4× over 20% random baseline) on real CKKS via metadata side-channels.
- Implemented Gaussian metadata noise-injection defense reducing attacker accuracy from **81%→71%** on TenSEAL and **91%→86%** on mock backend, with logistic-regression workloads remaining most distinguishable (F1=0.93).
- Identified noise budget proxy (0.33), execution time (0.27), and operation count (0.26) as top leakage channels via Random Forest feature importance, exposing how CKKS cryptographic confidentiality fails to protect operational privacy.

Wildfire Monitor — Real-Time Risk Early Warning System using Causal Time-Series ML Jan 2026 – Apr 2026

Next.js (TypeScript), D3.js, Python, scikit-learn, ARIMA, NASA FIRMS / Open-Meteo / Google Earth Engine APIs

- Designed Causal-Temporal Risk Decomposition (CT-RD) framework splitting wildfire risk into 5 interpretable components (ignition, spread, fuel, containment, impact) with dynamic environmental weighting, applying Causal ML and Explainable AI to wildfire forecasting.
- Developed 3-model ensemble forecaster (ARIMA + Gradient Boosting + Exponential Smoothing) across 3 horizons (24h/72h/7-day), achieving **0.041 MAE and F1=0.80** on alert classification via walk-forward backtesting on the Algerian Forest Fires dataset.
- Integrated 3 real-time data sources (Open-Meteo, NASA FIRMS, Google Earth Engine) into the feature vector pre-forecast via a live-context injection pipeline; ablation study showed cumulative **16.7% MAE reduction** over baseline when all live signals + ARIMA ensemble were enabled.

Agentic AI Support System — Multi-Agent LLM System for E-Commerce Migration Jan 2026

Python, LangChain, LangGraph, scikit-learn, sentence-transformers, Groq API, Streamlit

- Trained hybrid ML classifier (DBSCAN clustering + logistic regression on sentence-transformer embeddings) achieving **75.1% accuracy on 200 support tickets** across 3 incident categories with 98% clustering coverage.
- Designed 4-agent LangGraph orchestration (Investigator → Researcher → Analyst → Responder) with Retrieval-Augmented Generation (RAG) over Groq LLM inference, routing confidence-scored responses to auto-send (≥80%), human-approval (60–79%), or escalation (<60%) workflows.
- Architected self-updating documentation loop where the Responder agent autonomously creates GitHub pull requests when documentation-gap diagnosis exceeds 30% or cross-merchant patterns emerge across ≥3 merchants, closing the support-knowledge gap without human intervention.

TEMPUS — AI-Powered Android App for Smart Campus Management Aug 2025 – Nov 2025

Java (Android Studio), Python/FastAPI, YOLOv8, OpenCV, Mistral AI API, NGINX, Figma

- Shipped a 4-module integrated Android campus app deployed end-to-end (Java frontend, Python/FastAPI backend on custom-hosted domain), serving as a multi-course capstone spanning OS, Image & Video Processing, Software Engineering, and AI sub-modules.
- Engineered timetable OCR pipeline (bilateral filtering, adaptive Gaussian thresholding, 40×1 / 1×40 kernel cell detection, contour extraction) using Mistral AI OCR after benchmarking against EasyOCR, PyTesseract, and TrOCR — selected for superior cell-level text accuracy on tabular timetables.
- Developed YOLOv8 lift crowd detection with polygonal ROI counting and 3-tier density classification (red/yellow/green), Figma SVG floor-plan navigation across 5 floors, and a multi-format searchable notes portal with hierarchical file system design.

SKILLS

Languages: Python, Java, C++, JavaScript/TypeScript

ML / DL Frameworks: PyTorch, scikit-learn, Hugging Face (sentence-transformers), OpenCV, YOLOv8, TenSEAL

ML Domains: Deep Learning, Computer Vision, Natural Language Processing, Time-Series Forecasting, Agentic AI, Retrieval-Augmented Generation (RAG), Privacy-Preserving ML, Causal ML, Explainable AI

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Jupyter

Frameworks & Tools: FastAPI, Flask, LangChain, LangGraph, Streamlit, Docker, Git/GitHub, Next.js, REST APIs

Cloud & APIs: Groq API, Mistral AI API, Google Earth Engine, NASA FIRMS, Open-Meteo

ACTIVITIES

Taqneeq Fest, MPSTME Oct 2024 – Oct 2025

Social Media & Content Writing Subhead; Technicals & Collabs Subhead | Mumbai, India

- Led teams of 8 and 12 executives across two roles; designed and built the official Cyber Cypher event website, ideated wireframes for the Taqneeq mobile app, and produced 15+ Instagram Reels including coverage of a guest tech-influencer.

Association for Computing Machinery (ACM), MPSTME Aug 2022 – May 2025

Creatives Executive → R&D Senior Executive → Social Media Subhead | Mumbai, India

- Progressed through 3 roles across 3 years, overseeing teams of up to 8 executives; produced 15+ Instagram Reels for ACM's verified handle, authored articles for the ACM Chronicles newsletter, and led R&D and editorial operations.

CERTIFICATIONS & AWARDS

Certifications: Supervised Machine Learning: Regression and Classification (Stanford & DeepLearning.AI, Coursera); Introduction to Genomic Technologies (Johns Hopkins University, Coursera); Introduction to Psychology (Yale University, Coursera).

Awards: Best Delegate, IntraMUN 9.0; 1st Position, Nexus 1.0 (Colloqium, MPSTME); Distinction, Hyphen Ideathon (GDSC, MPSTME).